Maitri

A friend who tells the family what to do now.

Speak it. See why. Call at once.





Team Decyphers

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recHERsion 2026 | Problem Statement 1: Women's Healthcare

11:40 pm

Six hours after the birth, the bleeding has not stopped.

Her husband holds the phone. Her mother-in-law says wait for morning. Nobody knows if this is normal.



The next ten minutes decide.

An illustrative scene, not a real person.

Contact is high. The decision is late.

25.2M

births a year in India

Secondary source

48.6%

of maternal deaths in one Indian hospital study involved a delay in deciding or seeking care

Verified, single study

1 in 3

women still has fewer than four antenatal visits

Secondary, derived

Three delays kill. Maitri works on the first two.

1 **Recognise the danger**

2 **Decide to go now**

3 Reach and receive care

Why today's options fail the family

	Paper card	Helpline	Symptom chatbot	Maitri
Works with no internet	✓	✗	✗	✓
Hindi: listens and answers aloud	✗	✓	–	✓
Tells the family the next step	–	✓	–	✓
Shows why, with its sources	✓	✗	✗	✓
Hands a summary to the ASHA or doctor	✗	✗	✗	✓

Typical examples and our own assessment from limited searching, not every product. A helpline can give advice in Hindi; it needs a phone, signal and a clear head.

Built for the three people in the room



The woman

Pregnant, or within 42 days of birth



ASHA or ANM

Checks one woman after another



The family

Decides if she goes now

About 36% of women do not own a phone (secondary), so ASHA and family modes come first. First users: ASHAs and families in a Hindi-speaking district with patchy signal. Only TAM is a sourced number; SAM and SOM are marked "to be sized" on purpose.

Size of the opportunity

TAM

25.2M births a year in India

Secondary source

SAM

Births in Hindi-first, ASHA-served states

To be sized from SRS data

SOM

One district pilot, ASHAs of one block

To be sized with the health mission

Meet Maitri

Say it. See why. Know what to do.

SAY IT

Speak or tap, in Hindi or English. Hear every question and result read aloud.

SEE WHY

Every result names the rule and the source. No AI decides.

KNOW WHAT TO DO

One RED sign ends the check. One call, one plan, one slip.

Five things only Maitri does, together

1. Voice in, voice out
2. The call comes first
3. It shows its work
4. From panic to a plan
5. Private and offline

USP 1

Voice in. Voice out.

Speech-to-text and text-to-speech, built for a family that cannot type in a crisis



Honest note: speech recognition needs internet on most phones. Spoken output uses the phone's own voice.

1 Speak

One tap, then say what is happening in Hindi or English. Speech-to-text turns it into the signs.

2 Check and edit

She sees what was heard and the signs found, and can remove a wrong one. A spoken "No" is confirmed first.

3 Hear it back

Listen and Slow read every question and result aloud. Text-to-speech, in Hindi or English.

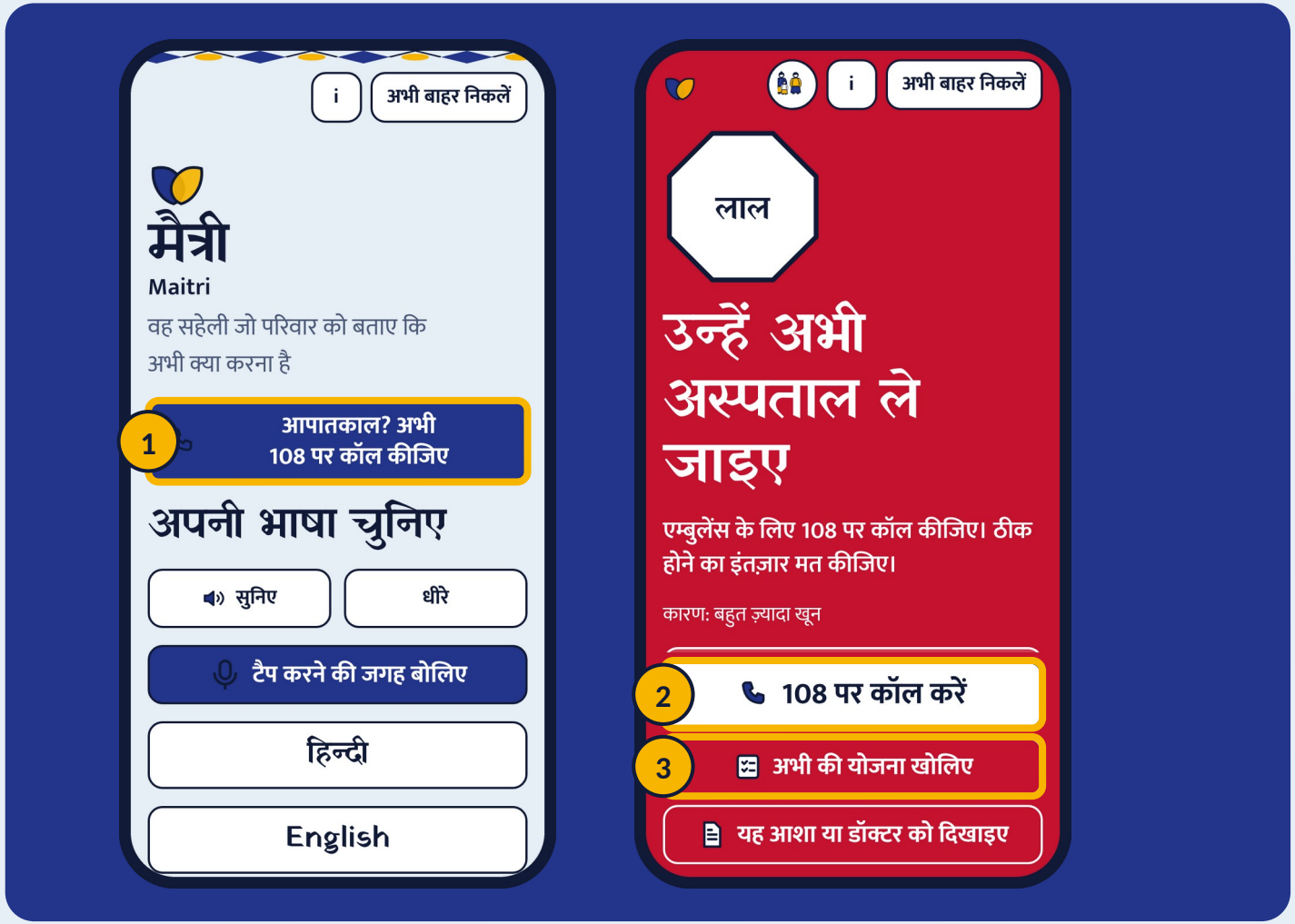
4 Answer by voice

Say yes, no or not sure on any question. Tapping always works too.

USP 2

The call comes first.

No account, no language choice, no questions between a family and the ambulance number



1 Emergency call is the first control

On the very first screen, in Hindi or English. The number comes from one config file.

2 One RED sign ends the check

No more questions. A large call button, then the go-now plan.

3 "Not sure" goes up. Never "fine".

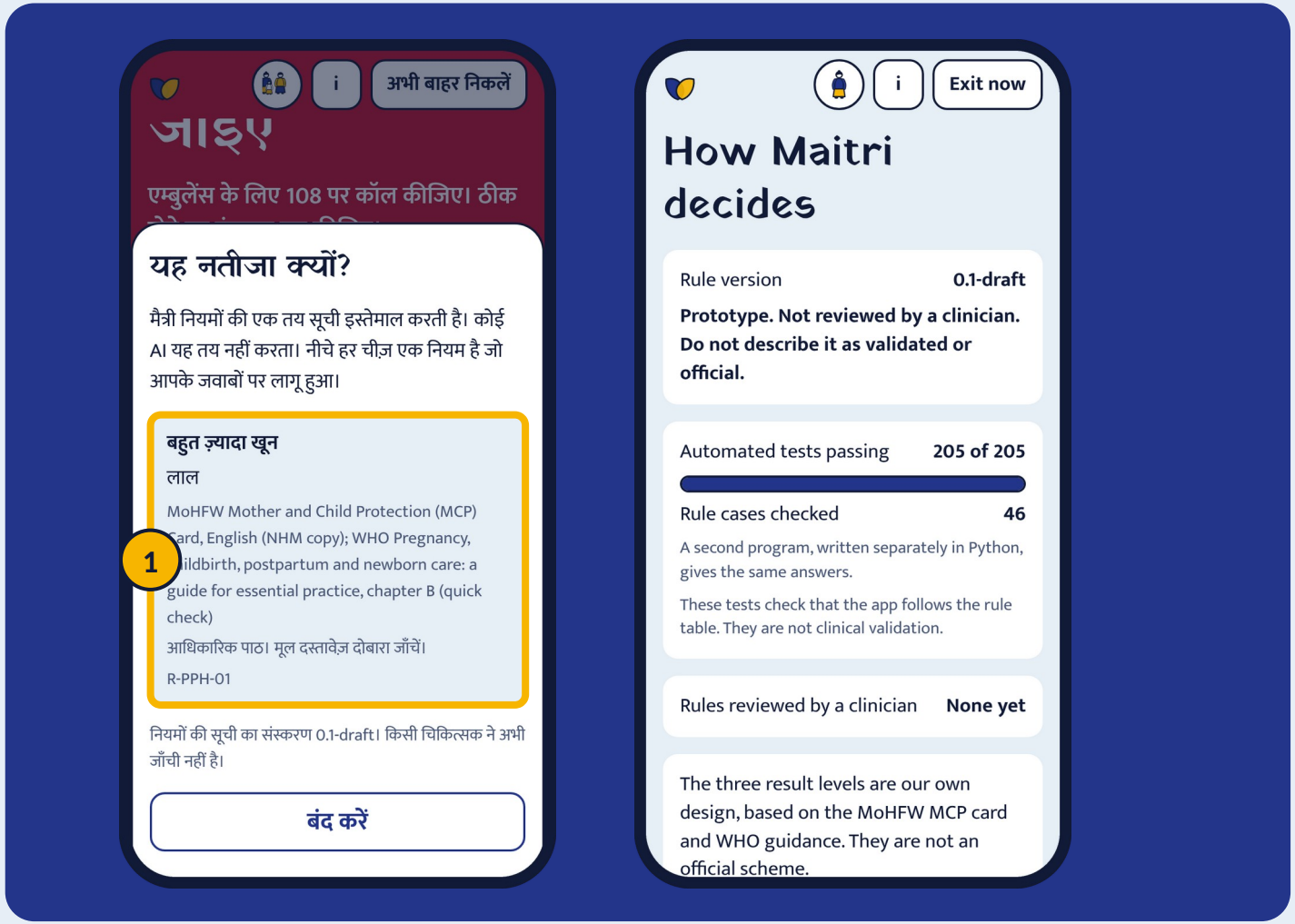
Over-escalating is a choice: missing a danger sign costs more than an extra visit.

- **Go to the hospital now**
- **See the ANM, ASHA or doctor today**
- **No danger sign told**

USP 3

It shows its work.

The whole reasoning on screen, in one tap, checkable by a clinician



1 **"Why this result?"**

The rule, the sign, the source, how reliable the source is, and the rule version.

2 **No AI decides**

A fixed list of rules in one file that a clinician can review without touching code.

3 **An honest Trust page**

Review status ("none yet"), helpline status, data handling and what is simulated.

205	46	12,420
automated tests	rule cases	cross-checked

USP 4

From panic to a plan.

The family leaves the screen knowing who to call, who goes and what to hand over



1 **The go-now plan**

Call, decide who goes, show the slip, ask for free transport and care (JSSK). One big call button.

2 **Steps to tick off**

Four steps in order, each in plain words, each with a Listen button.

3 **The Maitri slip**

Stage, result, signs told, care contacts with their status. No name, no phone number.

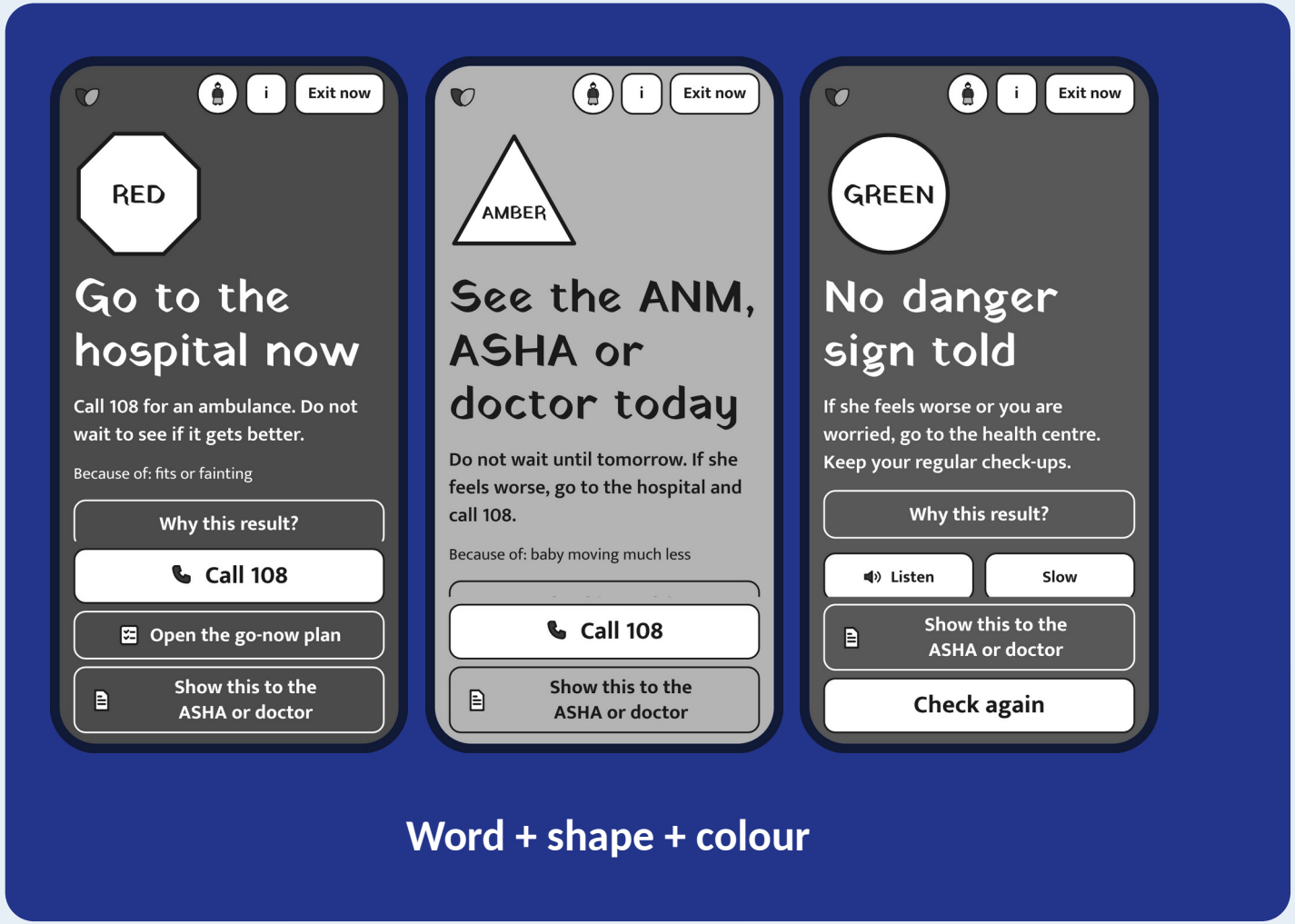
4 **Shared only with consent**

The ASHA reads it instead of asking again. She chooses a picture or plain text first.

USP 5

Private, offline, for everyone.

It still reads in greyscale, in bright sun, on a cheap phone, with no signal



Works offline

Verified in a browser with the network cut. Installs like an app.

Private by default

No account needed. "Exit now" clears everything and leaves.

Hindi and English, audited

A test fails if any screen shows the wrong language.

Pictures and big buttons

One question per screen, tick and cross, large targets, axe audit.

See it work, end to end



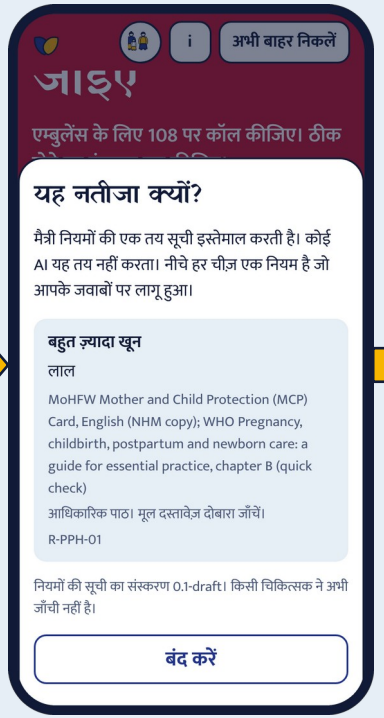
Emergency call first



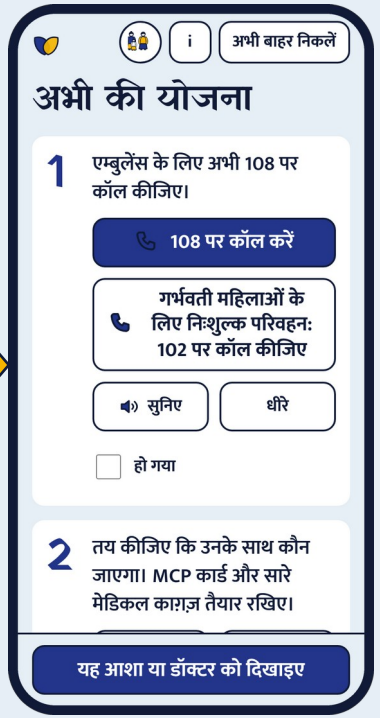
Say it, check it



One RED sign, one call



Why this result



Plan and slip

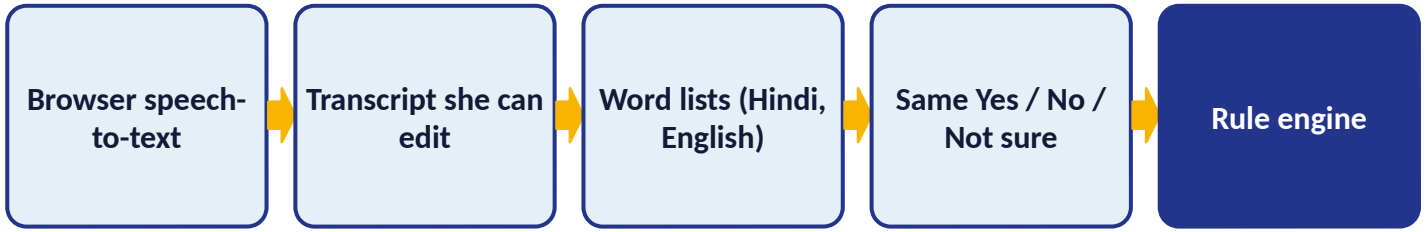
Code, video, live link

github.com/aafreenzy/maitri

About 30 seconds of video in docs/demo. The video's voice step uses a simulated recogniser; on a phone it is the browser's own.

How it is built

The voice path: no model decides



On the phone

- React 18 + TypeScript + Vite
- Rules in one JSON file
- Service worker: offline
- Hindi and English strings
- Speech-to-text and text-to-speech

Checks

- 205 automated tests
- Python re-check, 12,420 cases
- Database tests in real Postgres
- axe and small-screen audits
- CI on every pull request

Optional cloud (Supabase)

Sign in with Google or a phone code

checkins

user_id, entry_date, phase, answers (json), tier, rules_version

profiles

user_id, stage, due_date, delivery_date, language

Row level security: each woman reads only her own rows. Tested in Postgres.

Built, not connected yet

Built to be trusted. Honest about the gaps.

205	46	12,420	15	0
tests pass	rule cases checked	cases re-checked in Python	screens audited at 4 sizes	clinician reviews yet

Works today

- Tap and voice check
- Speech-to-text and text-to-speech
- RED, plan, slip, Why
- Offline, Hindi and English
- Trust page

Needs a real test

- Voice on real phones
- Sign-in and daily log
- Sharing on real phones

Not done yet

- Clinician review
- Recorded voice clips
- Field test with ASHAs
- More languages

Feasible now. Clear next steps.

Static site
No server for the check

Offline
Verified with the network cut

Low cost
About INR 6 per language for voice clips

Privacy first
No account needed

Now
Clinician review. ASHA field test. Real-phone voice test. Recorded voice clips.

Next
Odia, Assamese, Bengali with native review. Connect sign-in and the daily log.

Then
Visit and vaccine schedules. Scheme help. Each with an official source.

Later
Facility finder from a verified dataset.

Free for her. Funded by those who run care.



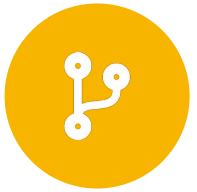
State missions and NGOs

Licence, localisation, ASHA training and a district dashboard (with consent)



CSR and grants

Fund new languages, voice clips and clinician review



Open core

Rules and app stay open and auditable. Services are paid.

We never sell data. No ads. Women and families pay nothing.

A proposed model, not yet tested with a funder.

What broke. What we changed.

The build failed on main

A new screen changed the flow and a test was not updated. Tests now gate the build and CI runs on every pull request.

Buttons clipped on short phones

Items were squeezed in a scrolling column. One layout rule and a 15-screen audit fixed it.

A misheard "No" could hide a sign

A spoken No is confirmed. A misheard Yes only over-escalates, so it goes straight through.

Voice cannot be fully tested here

Tests use a simulated recogniser, and we say so. A real-phone test is the next step.

Trade-offs we chose

Rules, not AI | **Over-escalate, never under** | **Offline over cloud** | **No numeric thresholds**

Every minute counts.

Maitri makes the next one clear.

Speak it. See why. Call at once.

To take it from prototype to pilot, we need three things

A clinician reviewer

To check and sign off the rules

An ASHA district

To pilot with real families

Language partners

Odia, Assamese, Bengali

Scan for code and video

Maitri. The friend who tells the family what to do now.

Education tool. Does not diagnose. Rules are a draft, not yet reviewed by a doctor.

Sources and credits

- MoHFW Mother and Child Protection card | HBNC leaflet | WHO essential practice guide, chapters B and D
- JSSK guidelines (NHM) | PIB note on Tele-MANAS 14416
- NFHS-6 findings and SRS maternal mortality bulletin 2021-23 (secondary coverage, labelled in the repository)
- BMC Pregnancy and Childbirth 2019 delay study | Cochrane 2020 mobile messaging review | Kenya PROMPTS trial
- Built with React, TypeScript, Vite, Vitest, PGlite (tests), Supabase client. Fonts: Yatra One and Mukta (SIL OFL). Pictograms drawn for this project.
- AI assistance: built with Claude Code. The team is responsible for all content.

github.com/aafreenzy/maitri

Education tool. Does not diagnose. Rules are a draft, not reviewed by a doctor.